

Azure AD in Hybrid joined

Components Required

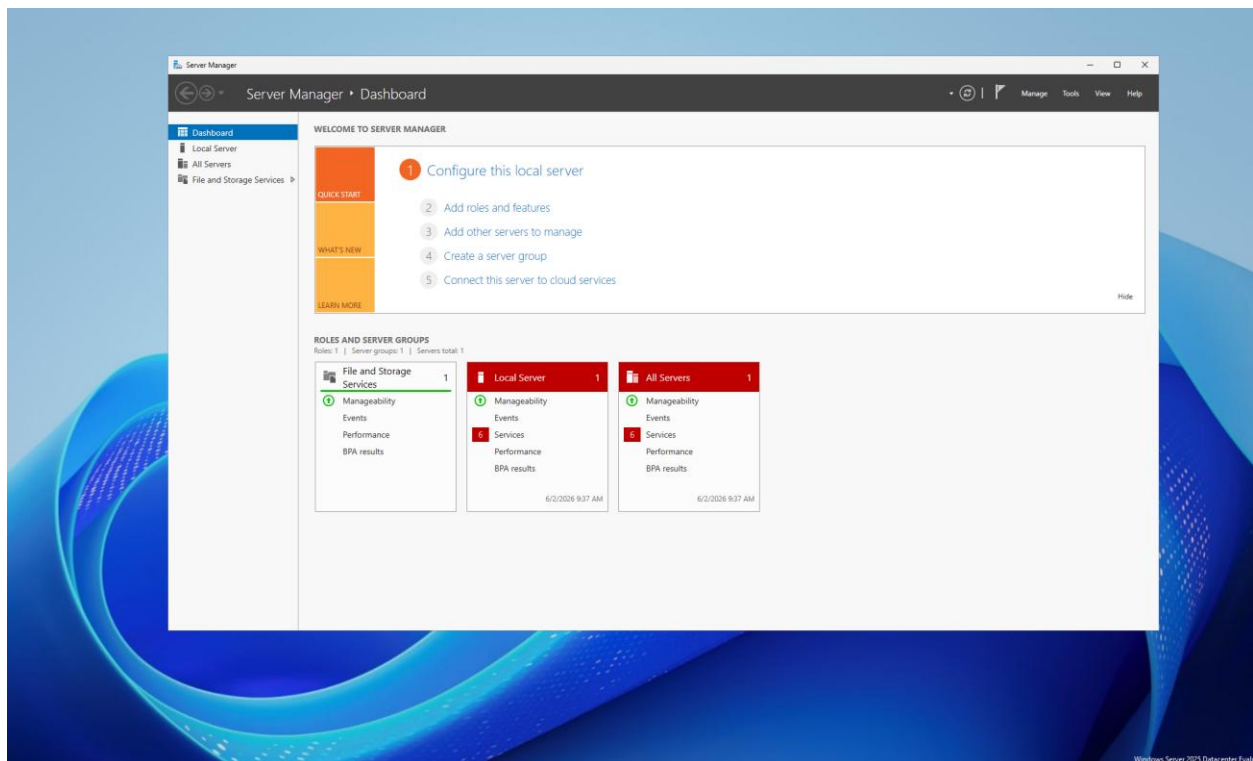
On-Premises

- Windows Server 2022 VM
- Active Directory Domain Services
- DNS Server
- Windows 11 Client VM

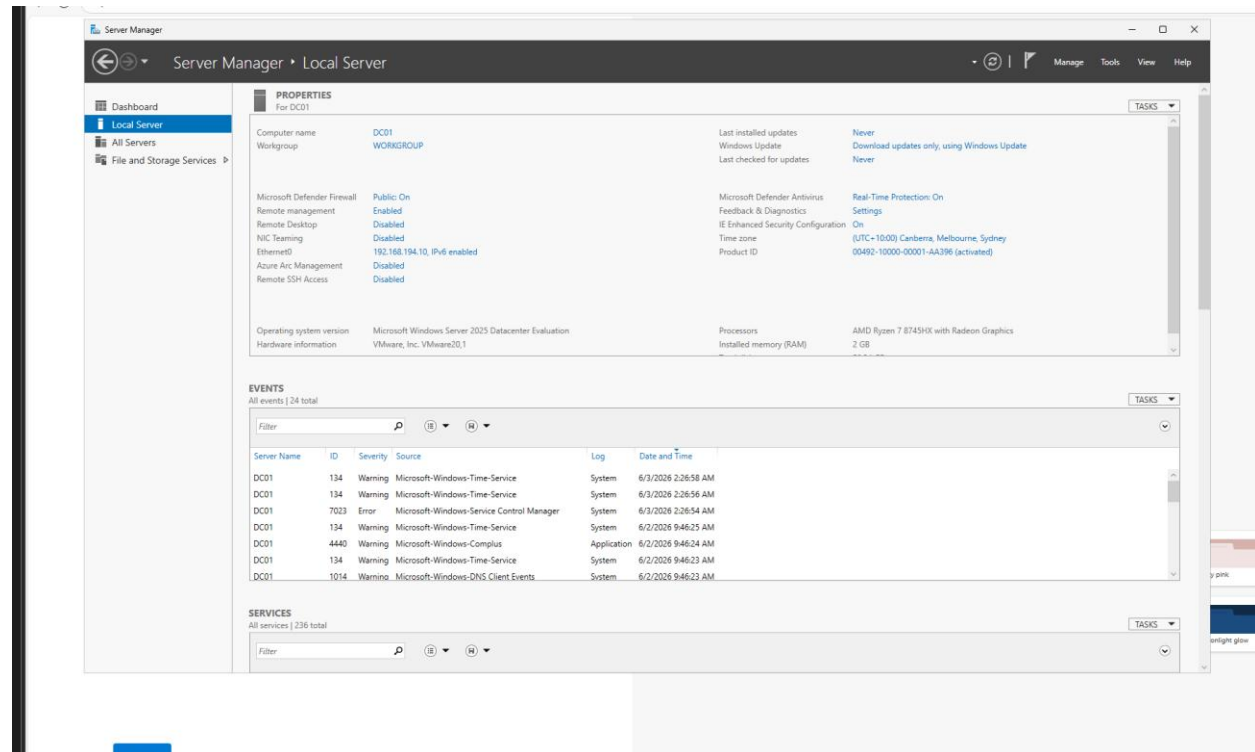
Cloud

- Azure Free Account
- Microsoft Entra ID
- Azure AD Connect
- Microsoft 365 Developer Tenant (Free)

Step 1: Install windows server 2025, and install active directory domain services on it.



And the server is getting IP address from the DHCP so make it static by going into network settings. And change the computer name to DC01. And restarted the system

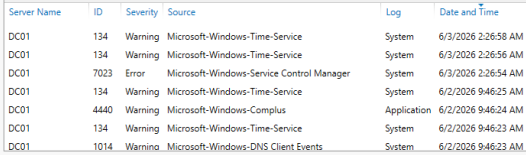


Install the active directory

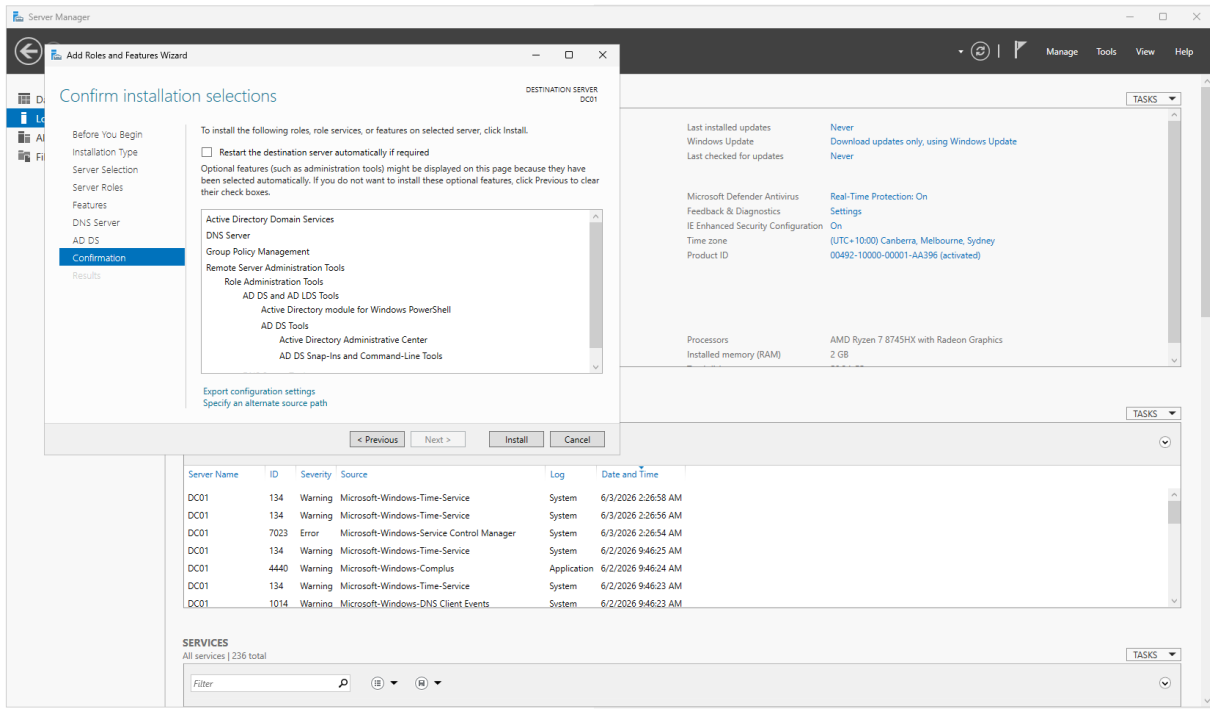
Go to Manage->Add Roles and Features-> and install Active Directory Domain Services and DNS Server



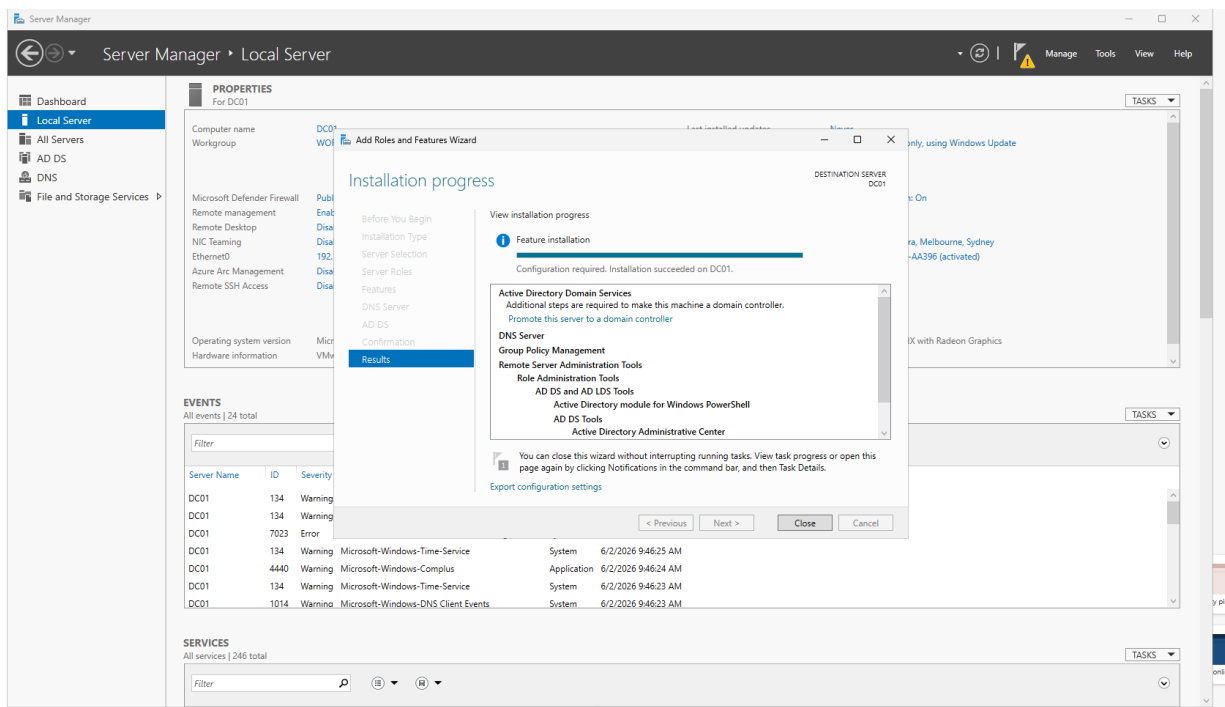
● ○



And verify installations and click on install

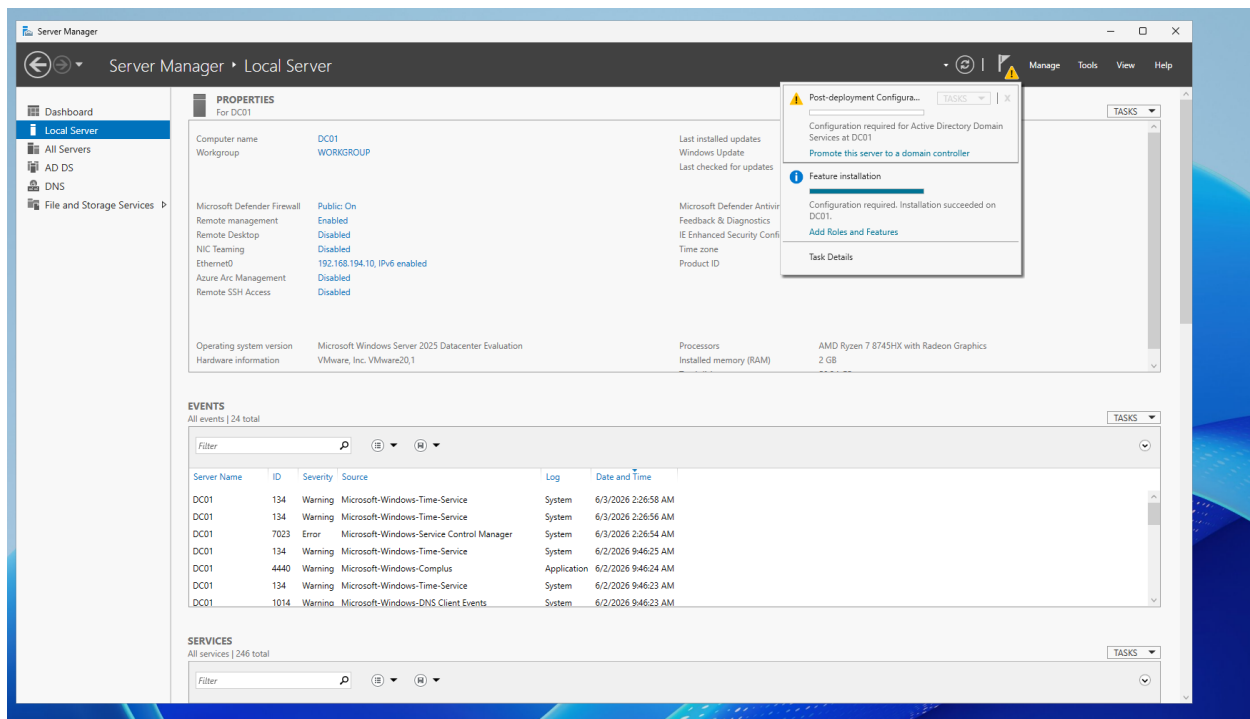


Active directory installation is completed. Close it

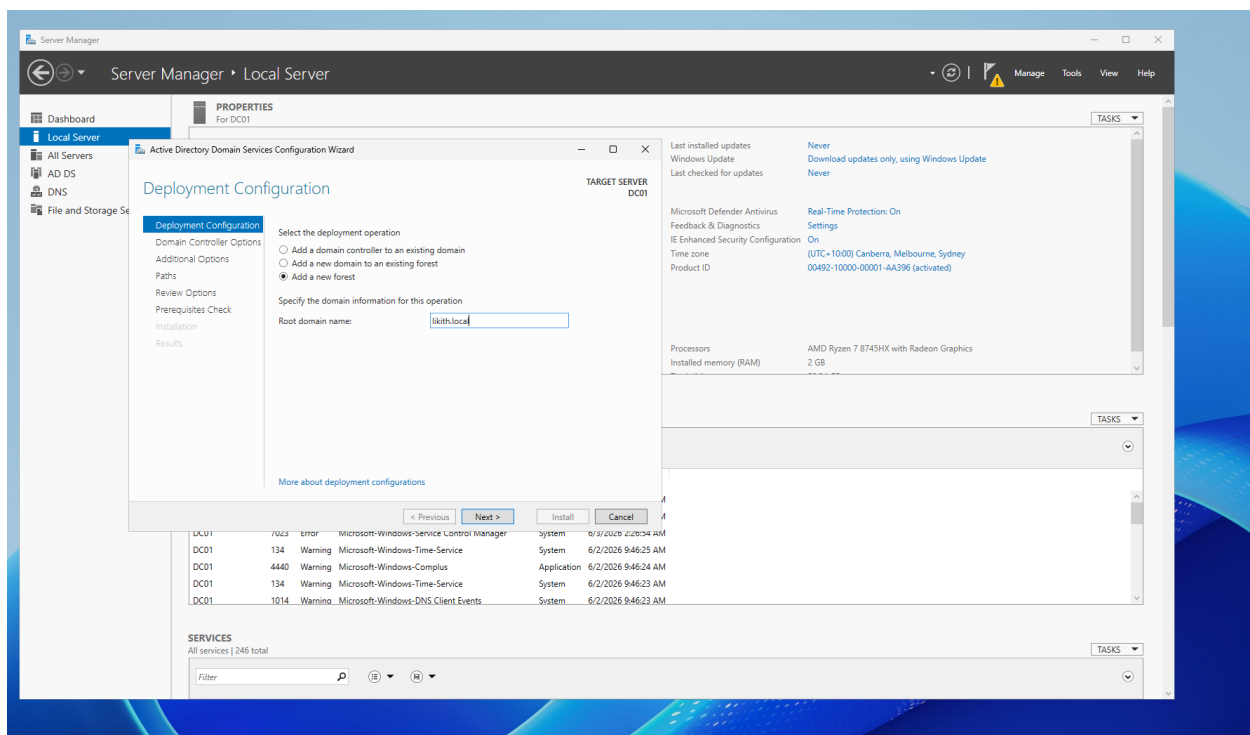


To promote the server to Domain controller, Select the notification ikon on left to manage.

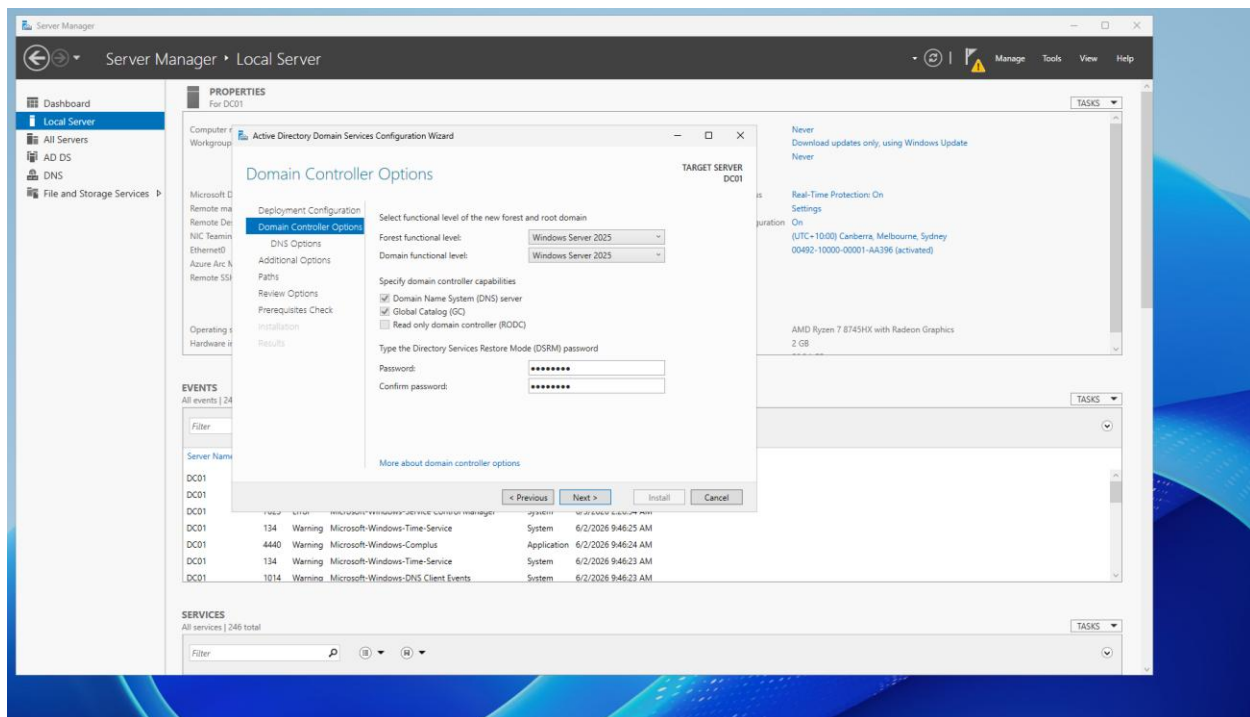
Click on promote this server to domain controller



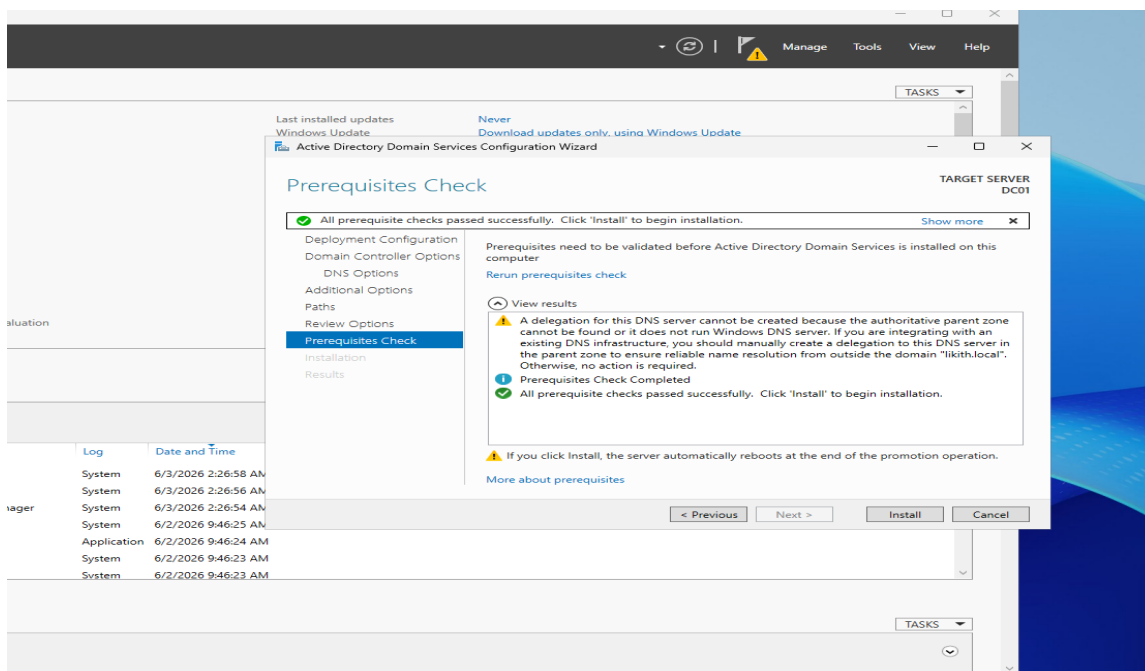
Since we don't have forest we need to create a new domain I have named it as likith.local



Select the forest function level and DSRM password

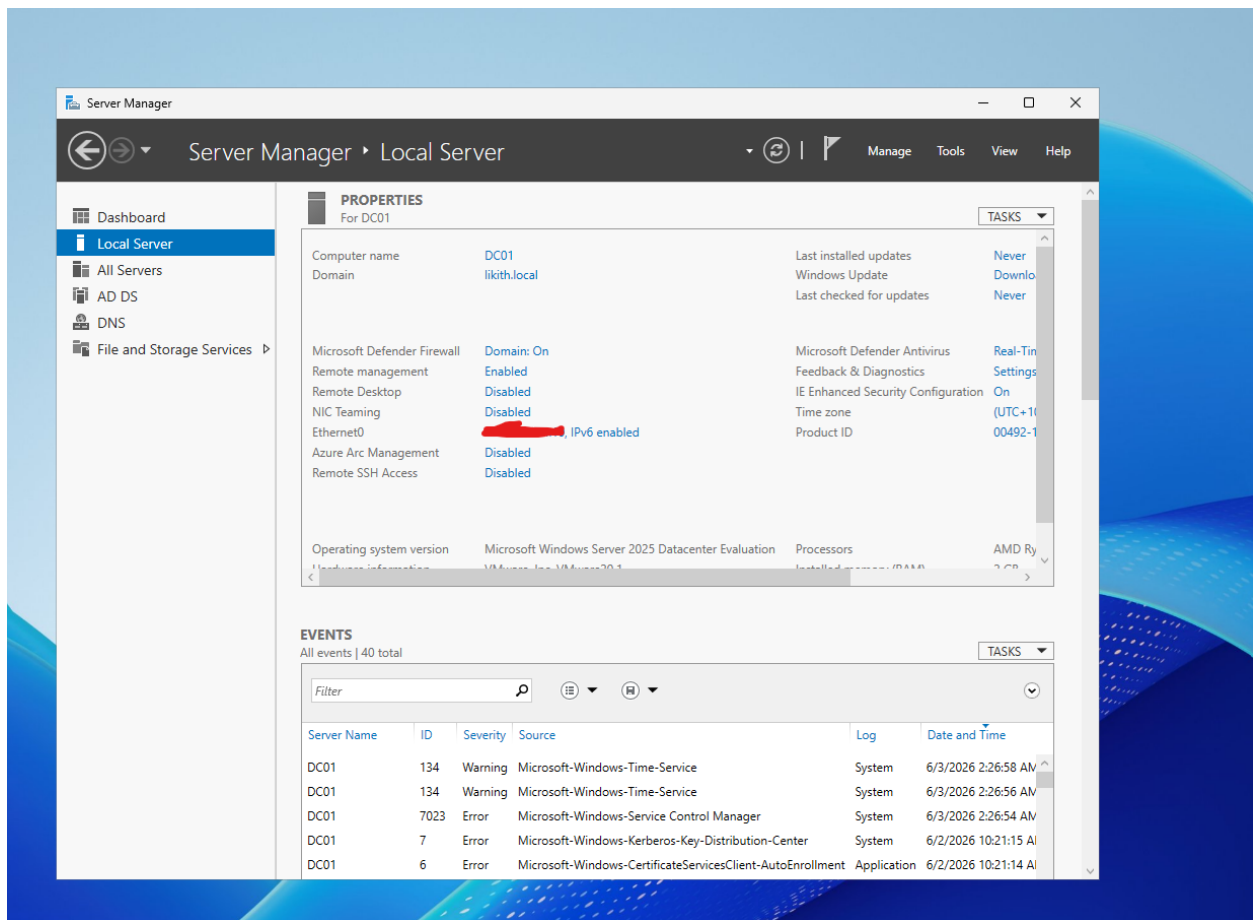


Run the prerequisite and check for errors and proceed to installation

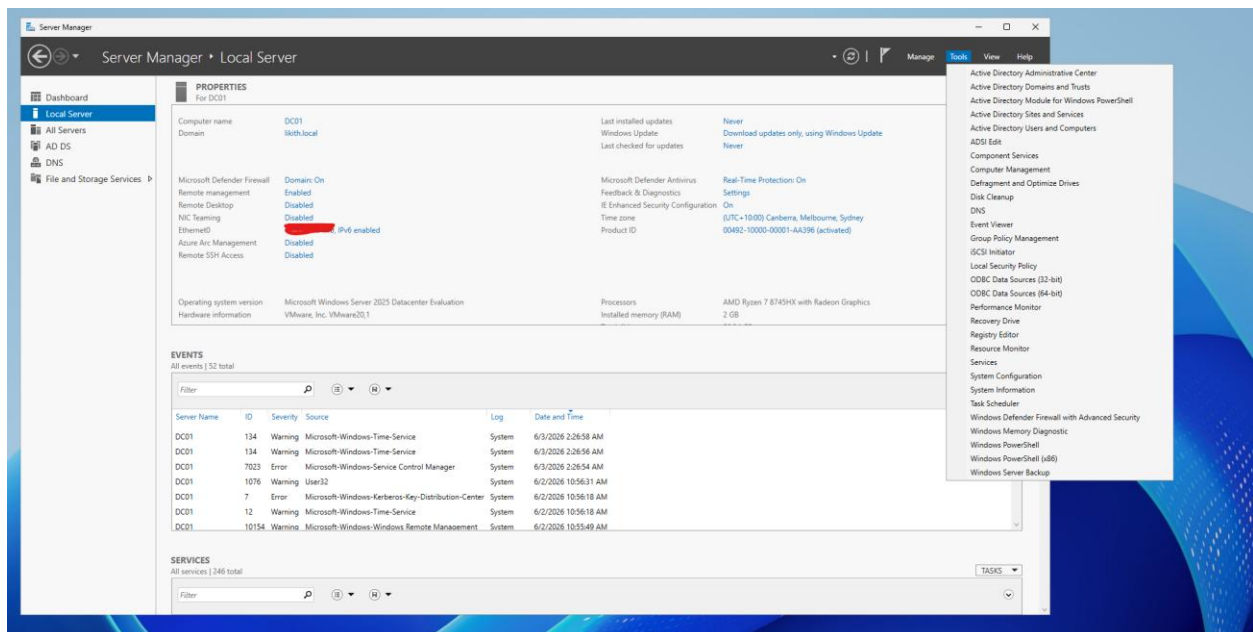


After Installation the server need to be rebooted

We can see the domain name has changed to likith.local and we can see AD DS is installed

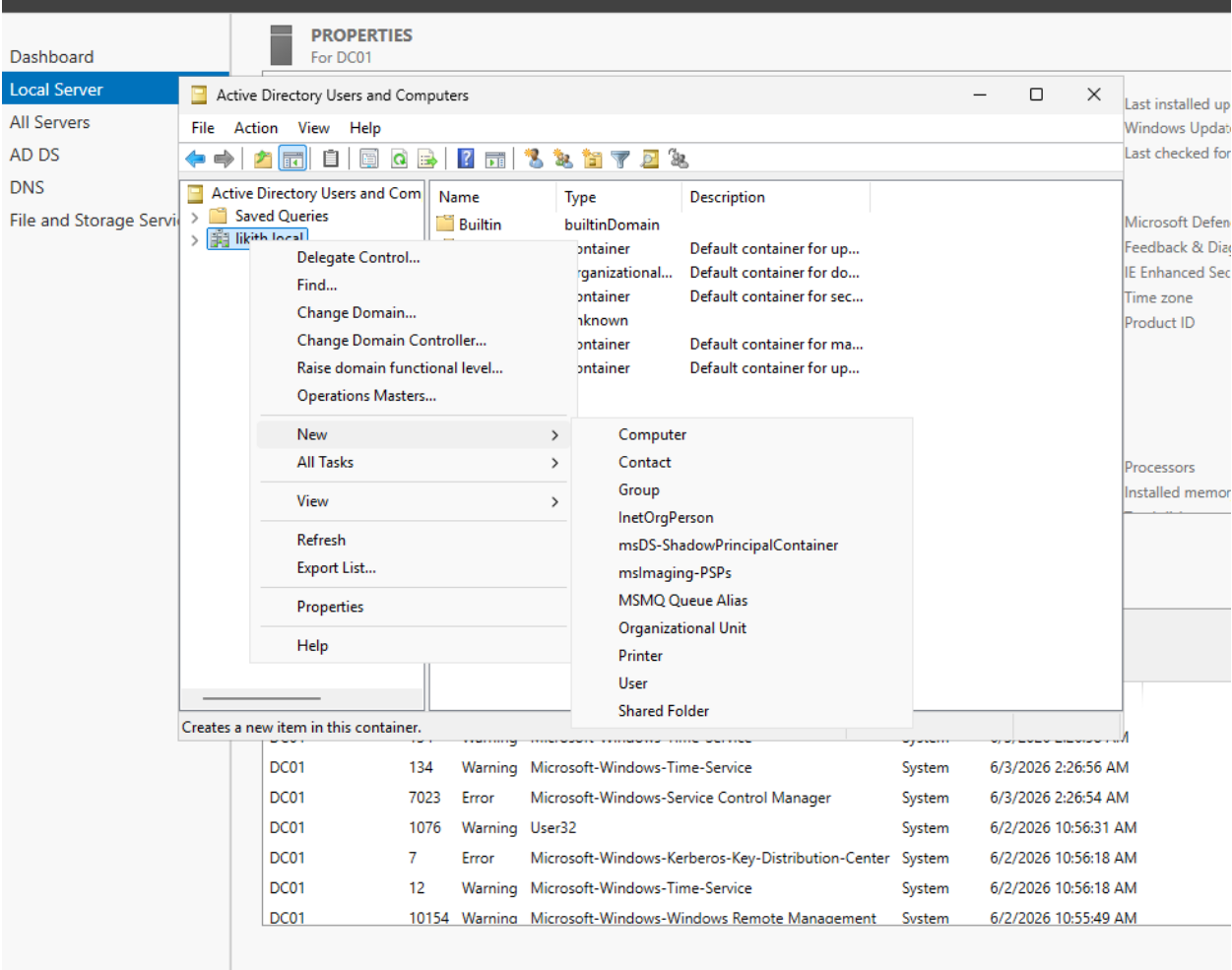


To **Create** Organizational Units Active Directory Users and Computers:



Created these OUs:

- Employees
 - IT
 - HR
 - Computers
 - Groups



Created 3 users

The screenshot displays the Windows Server 2016 interface. The 'Local Server' dashboard is visible on the left. The 'Active Directory Users and Computers' window is open, showing a list of users and groups. The 'Users' folder is selected, and the list includes 'Likith Admin' and 'Test User'. The 'Event Viewer' window is also open, showing a list of events. The 'System' log is selected, and the list includes events from 'Microsoft-Windows-Time-Service' and 'Microsoft-Windows-Service Control Manager'. The 'System properties' window is open, showing the 'Hardware' tab with details about the processors and memory.

Name	Type	Description
Domain Admins	Security Group...	Designated administrators of this domain
Domain Controllers	Security Group...	All domain controllers in this domain
Domain Guests	Security Group...	All domain guests
Domain Users	Security Group...	All domain users
Enterprise Administrators	Security Group...	Designated administrators of this domain
Enterprise Key Administrators	Security Group...	Members of this group can manage cryptographic keys
Enterprise Read-Only Administrators	Security Group...	Members of this group can manage cryptographic keys
External Trusted Accounts	Security Group...	All external trust accounts in this domain
Forest Trust Built-in Group	Security Group...	All forest trust accounts in this domain
Group Policy Objects	Security Group...	Members in this group can manage group policy objects
Guest	User	Built-in account for guest users
Key Administrators	Security Group...	Members of this group can manage cryptographic keys
Likith	User	
Likith Admin	User	
Protected Users	Security Group...	Members of this group can manage cryptographic keys
RAS and IAS Servers	Security Group...	Servers in this group can manage remote access
Read-only Domain Controllers	Security Group...	Members of this group can manage cryptographic keys
Schema Administrators	Security Group...	Designated administrators of this domain
Test User	User	

ID	Source	Category	Message	Level	Time
DC01	134	Warning	Microsoft-Windows-Time-Service	System	6/3/2026 2:26:56 AM
DC01	7023	Error	Microsoft-Windows-Service Control Manager	System	6/3/2026 2:26:54 AM
DC01	1076	Warning	User32	System	6/2/2026 10:56:31 AM
DC01	7	Error	Microsoft-Windows-Kerberos-Key-Distribution-Center	System	6/2/2026 10:56:18 AM
DC01	12	Warning	Microsoft-Windows-Time-Service	System	6/2/2026 10:56:18 AM
DC01	10154	Warning	Microsoft-Windows-Windows Remote Management	System	6/2/2026 10:55:49 AM

SERVICES
All services | 246 total

Filter

Create 3 Groups for HR,IT and managers

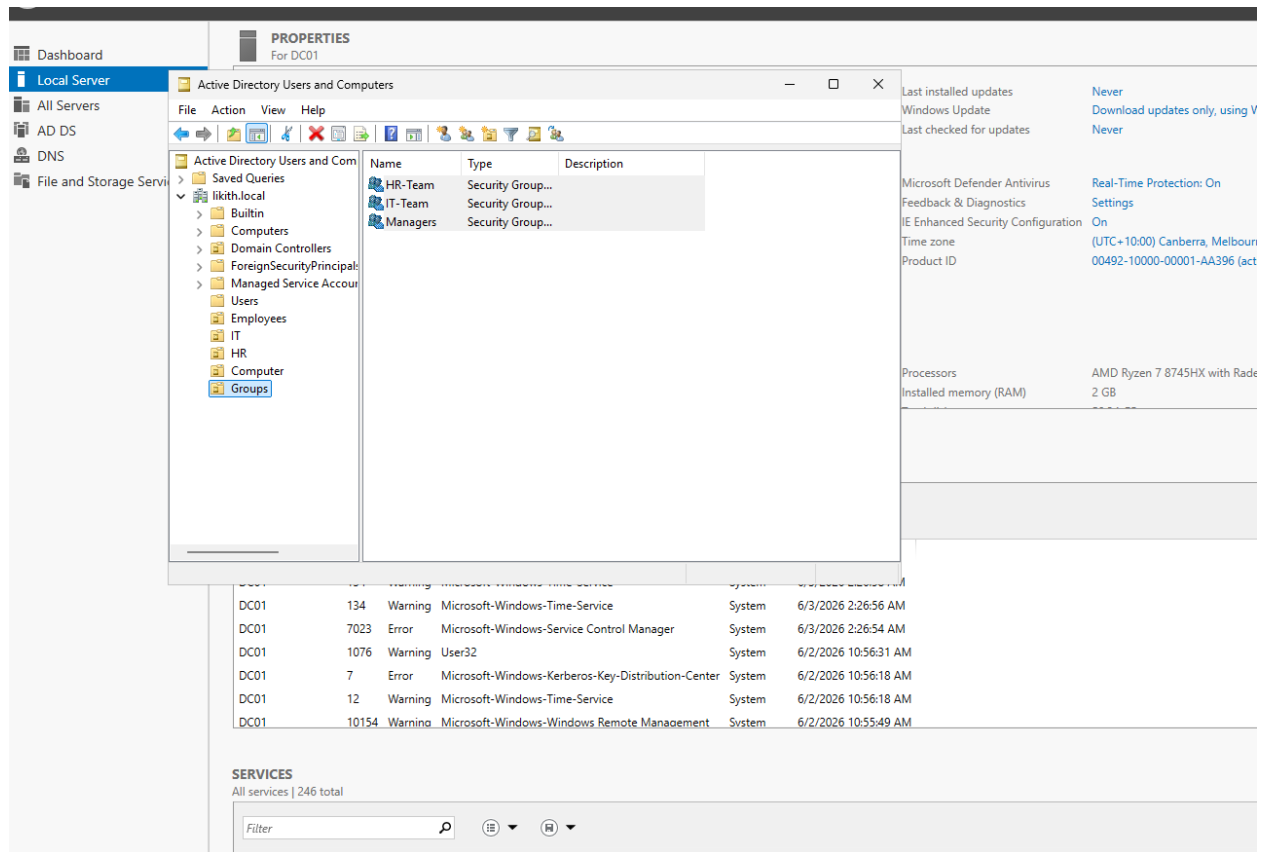
Right-click Groups → New → Group

Group name: HR-Team

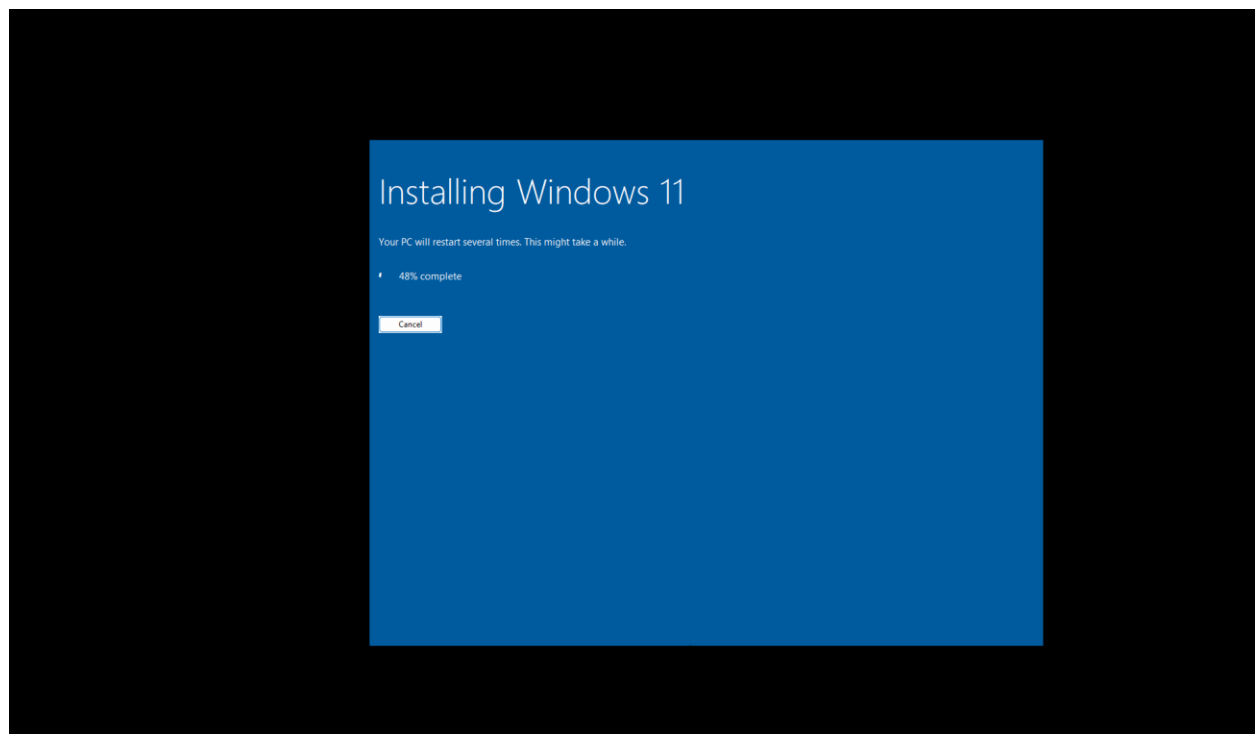
Group scope: Global

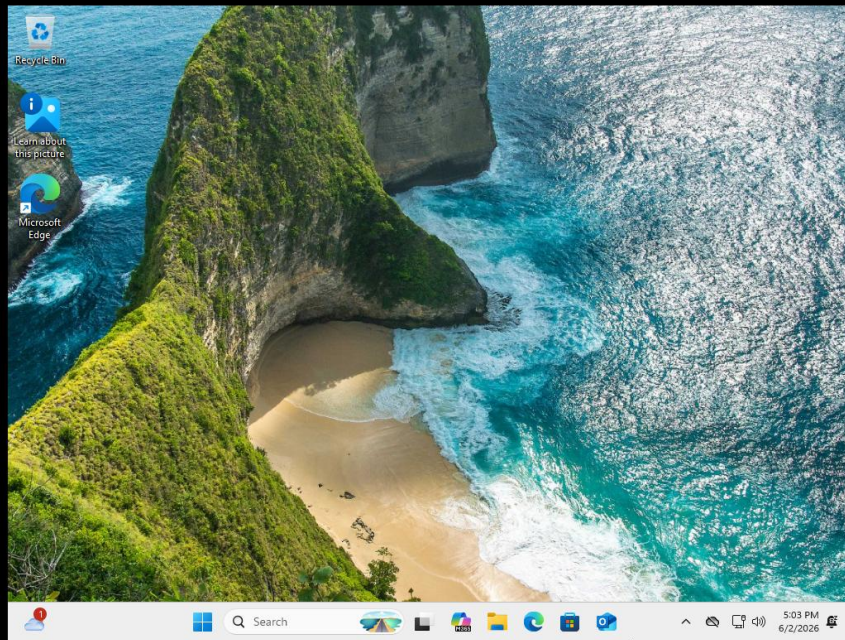
Group type: Security

Add the user to the groups



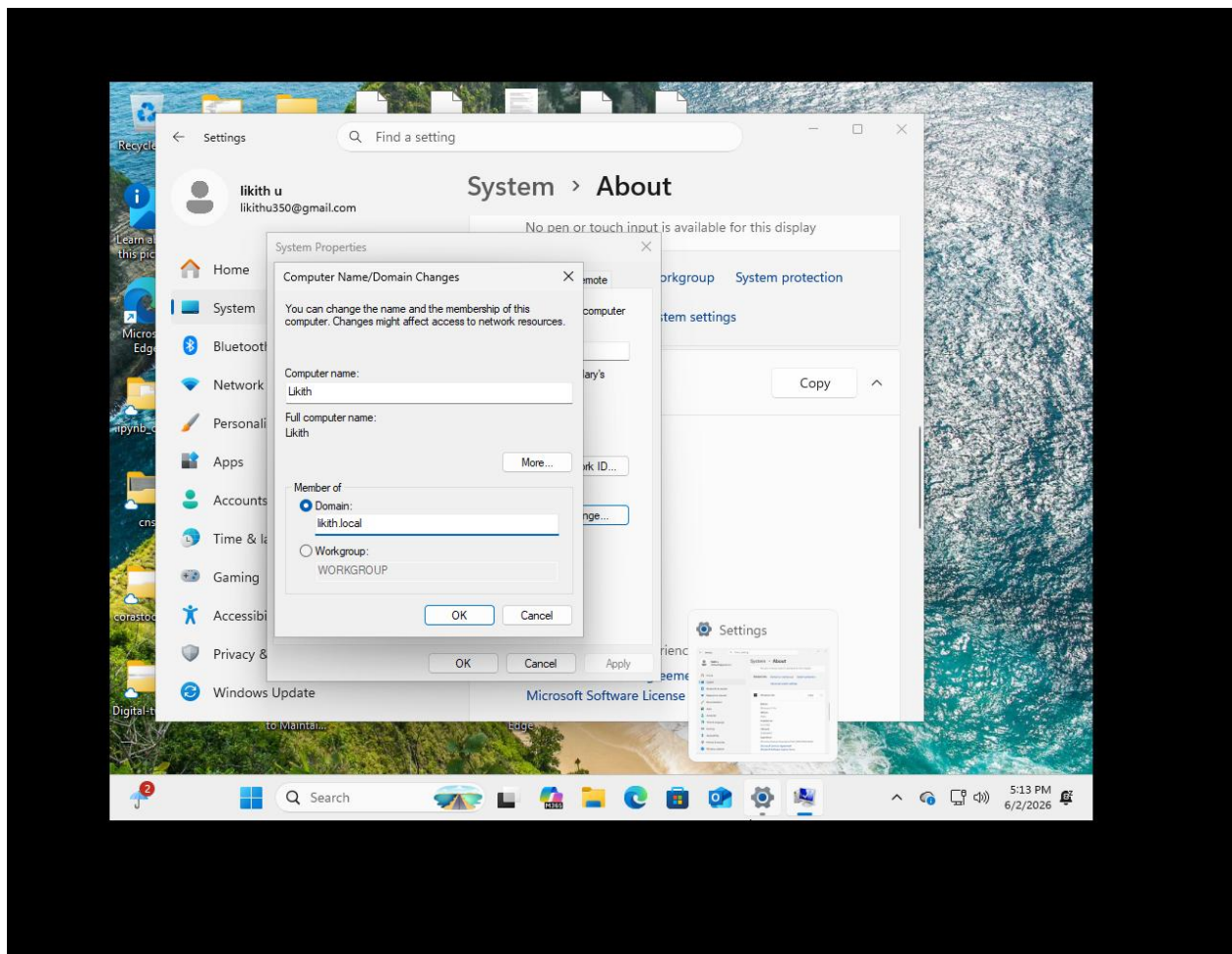
Step 2: Install Windows 11 operating system on the VMware





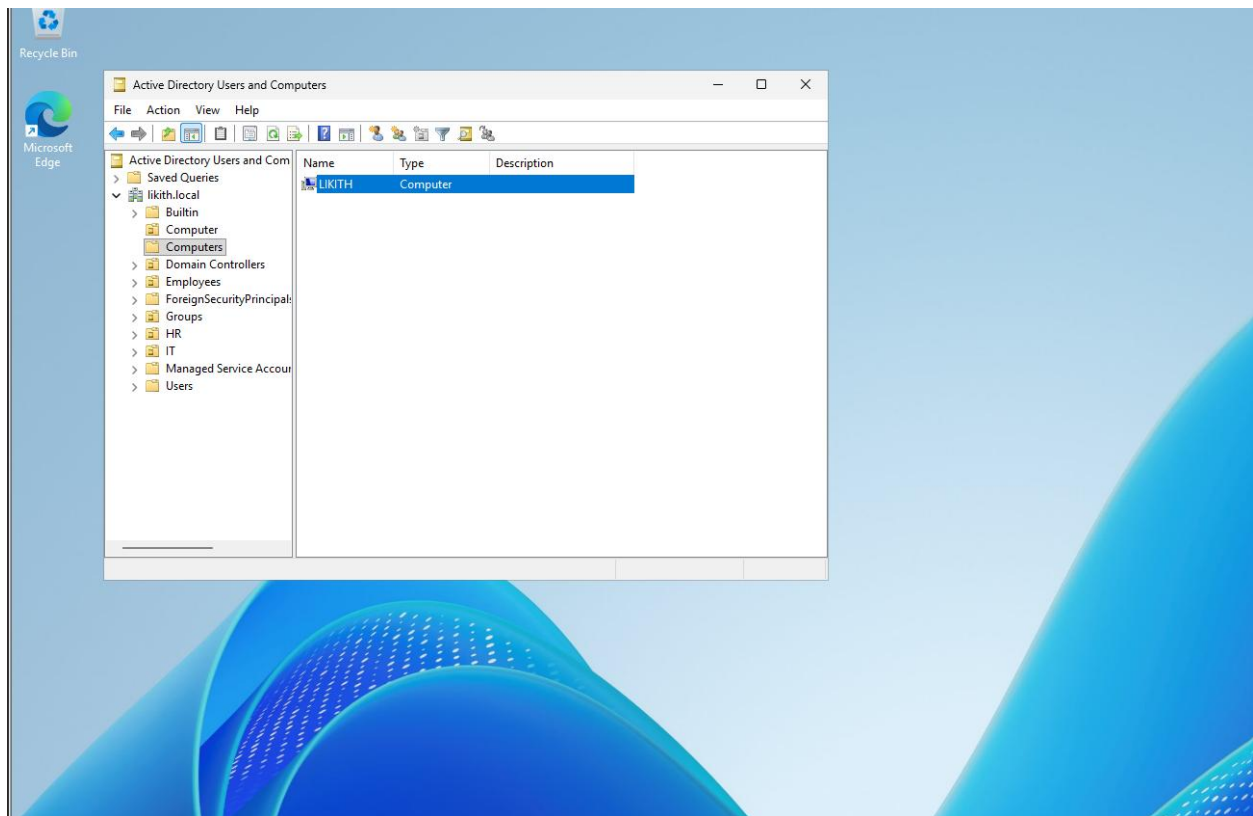
Assign the static IP address for the windows PC and check if the domain controller is accessible by pinging the IP address of the domain controller.

Joining the PC to domain controller



Restart the PC to reflect

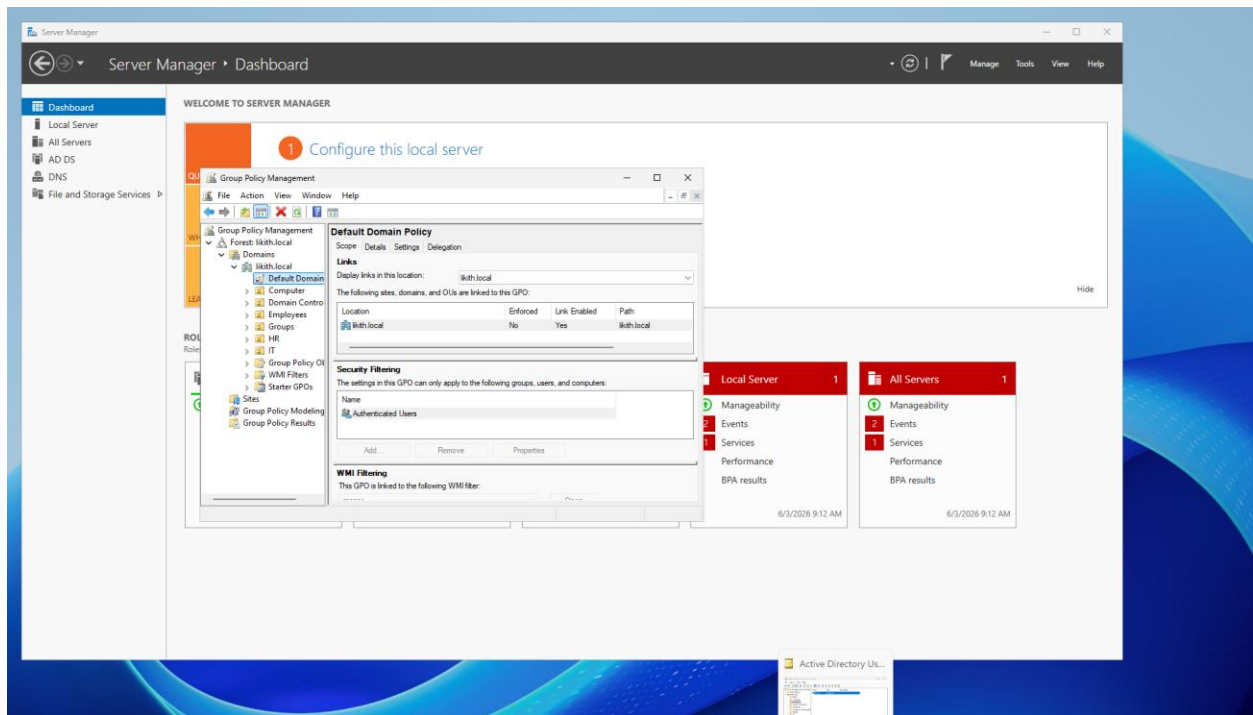
Go to the active directory to verify the computer is joined to domain controller or not.



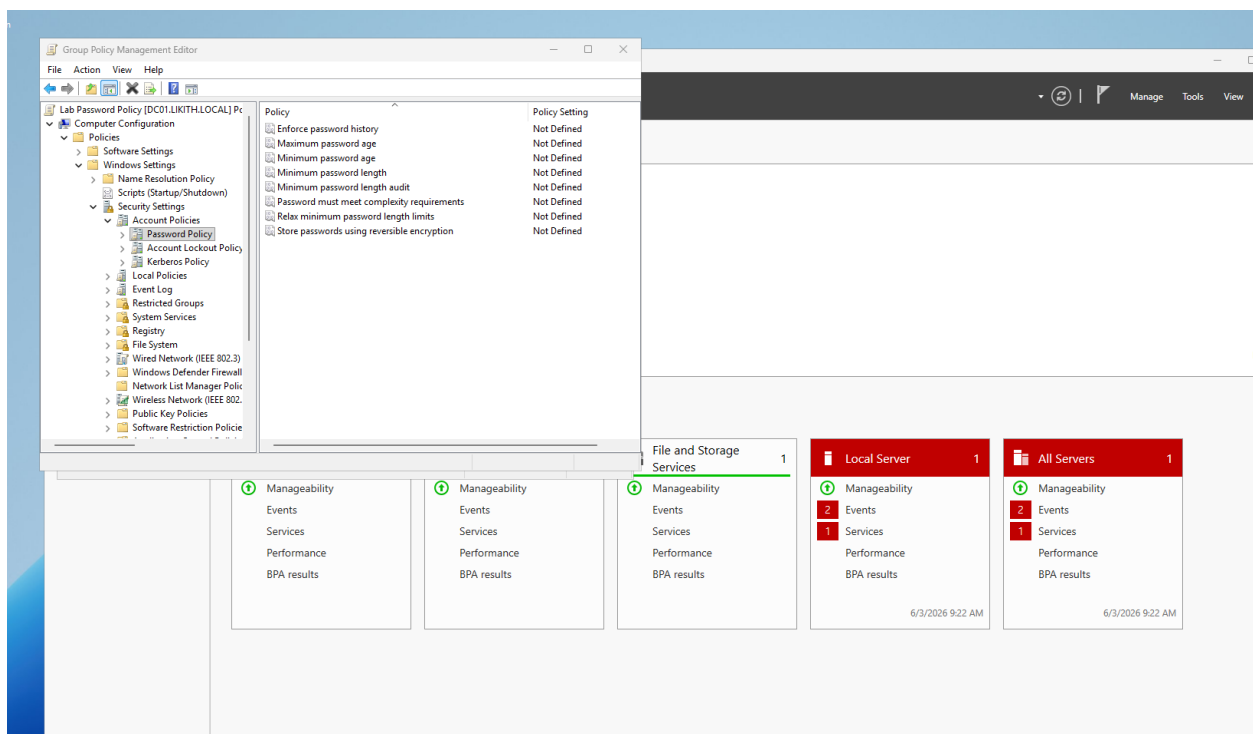
Creating a group policy:

Create GPO: for Password policy, desktop wallpaper and USB restriction

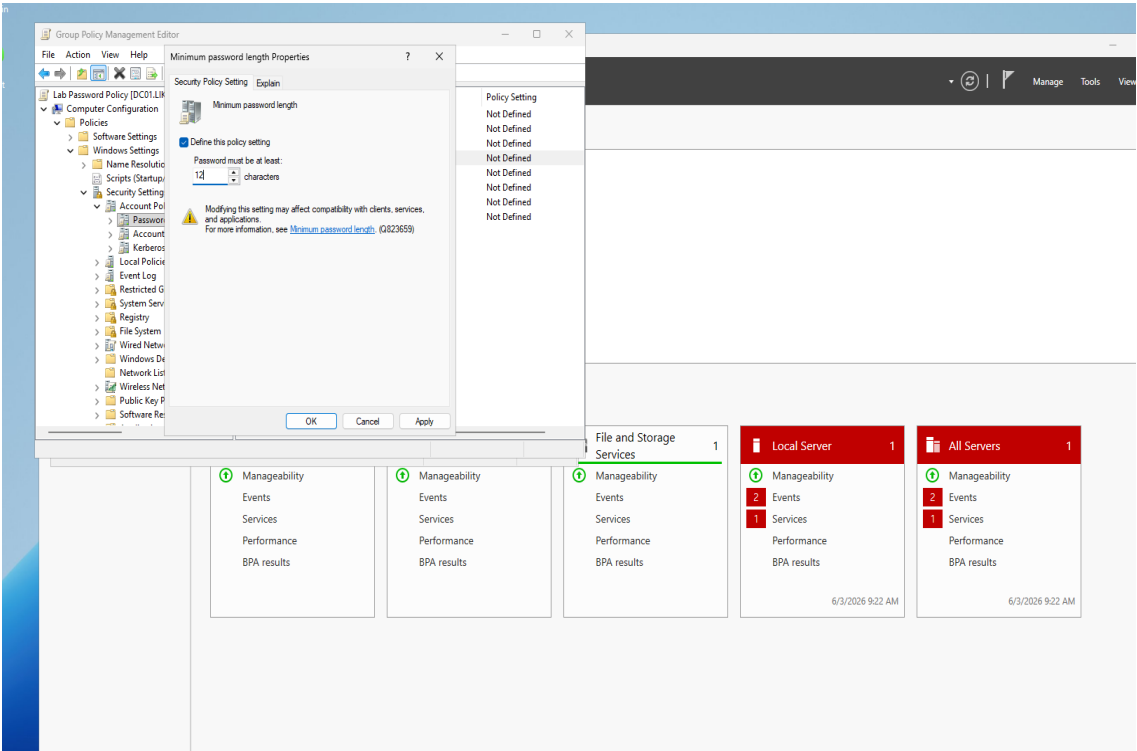
To do open group policy management Forest and right click on Default Domain Policy and select edit.



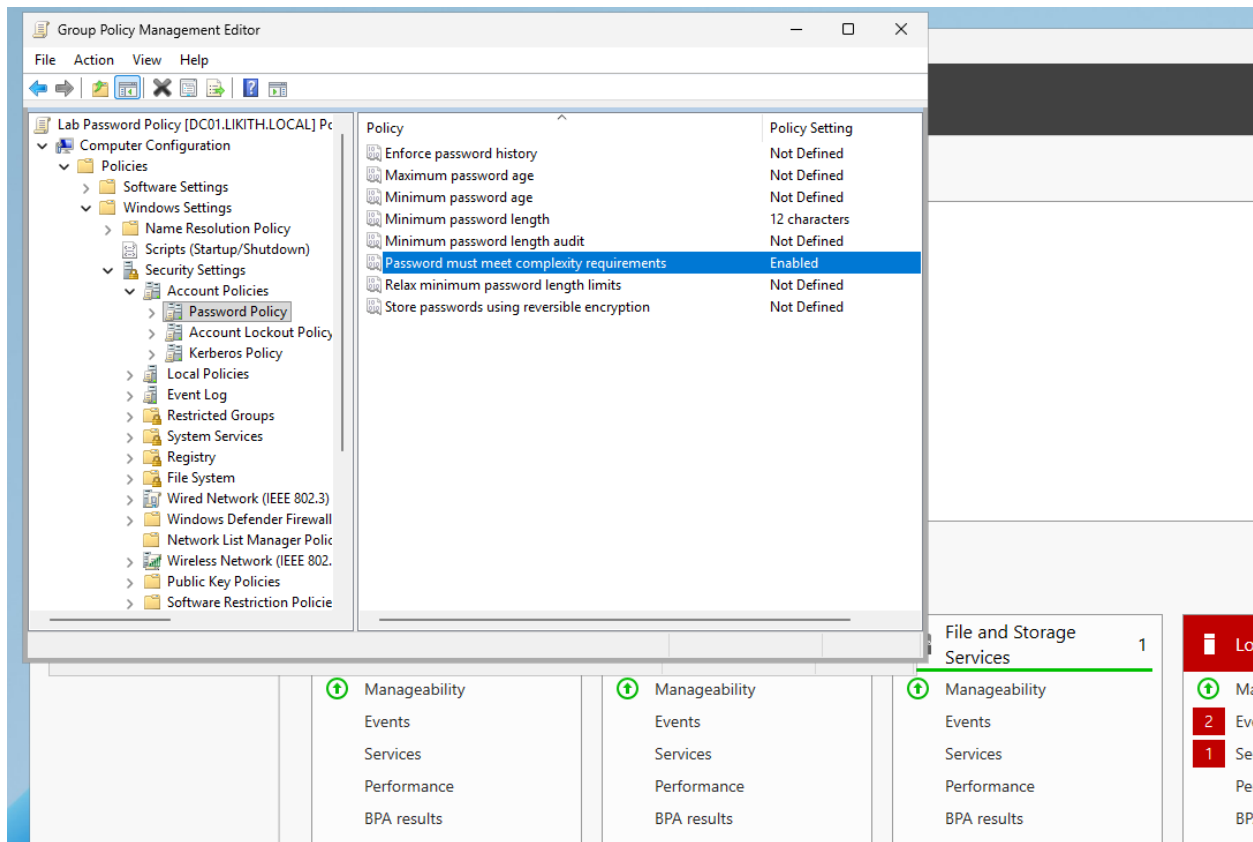
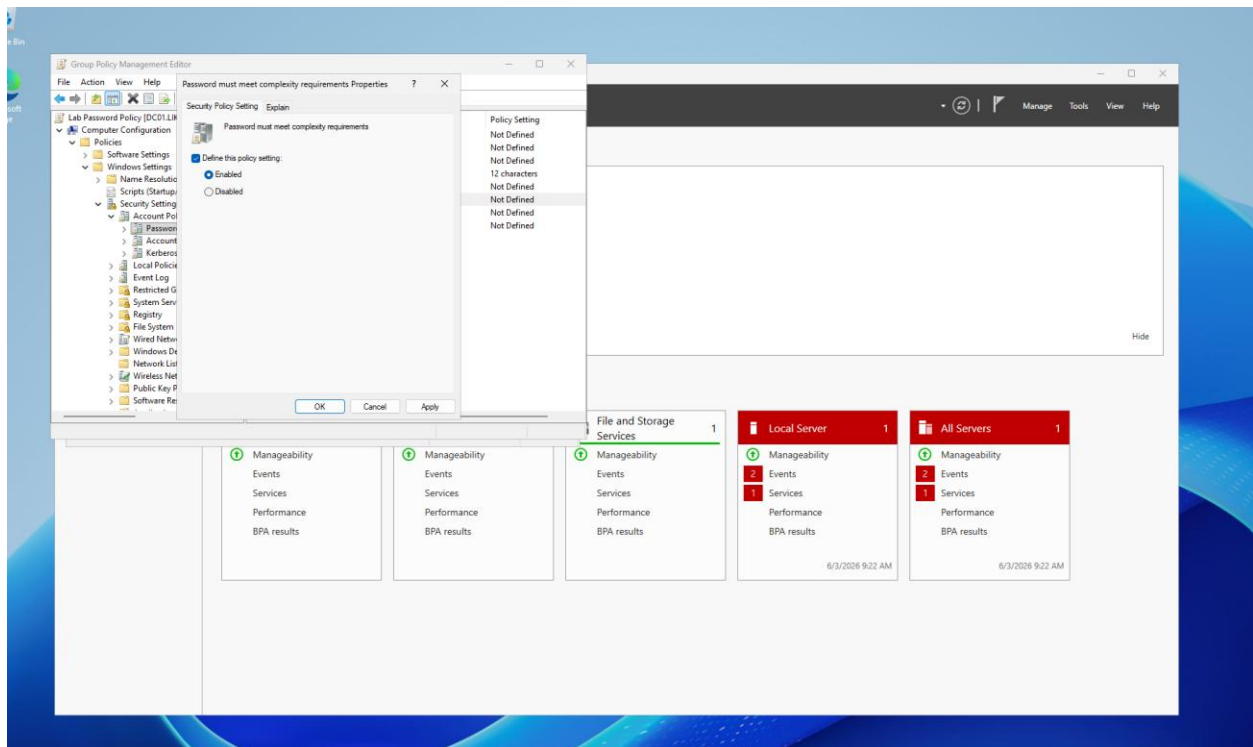
- Password Policy
Minimum password length = 12
Password complexity = Enabled



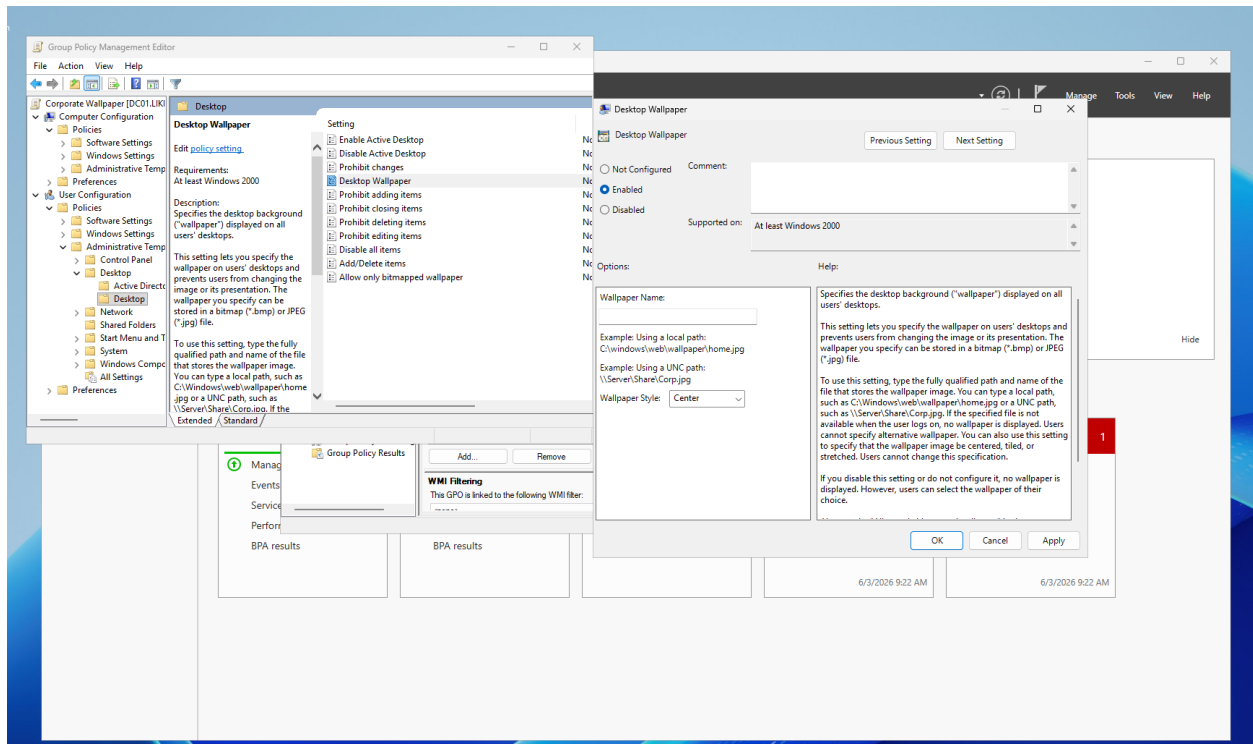
Change the password length to 12



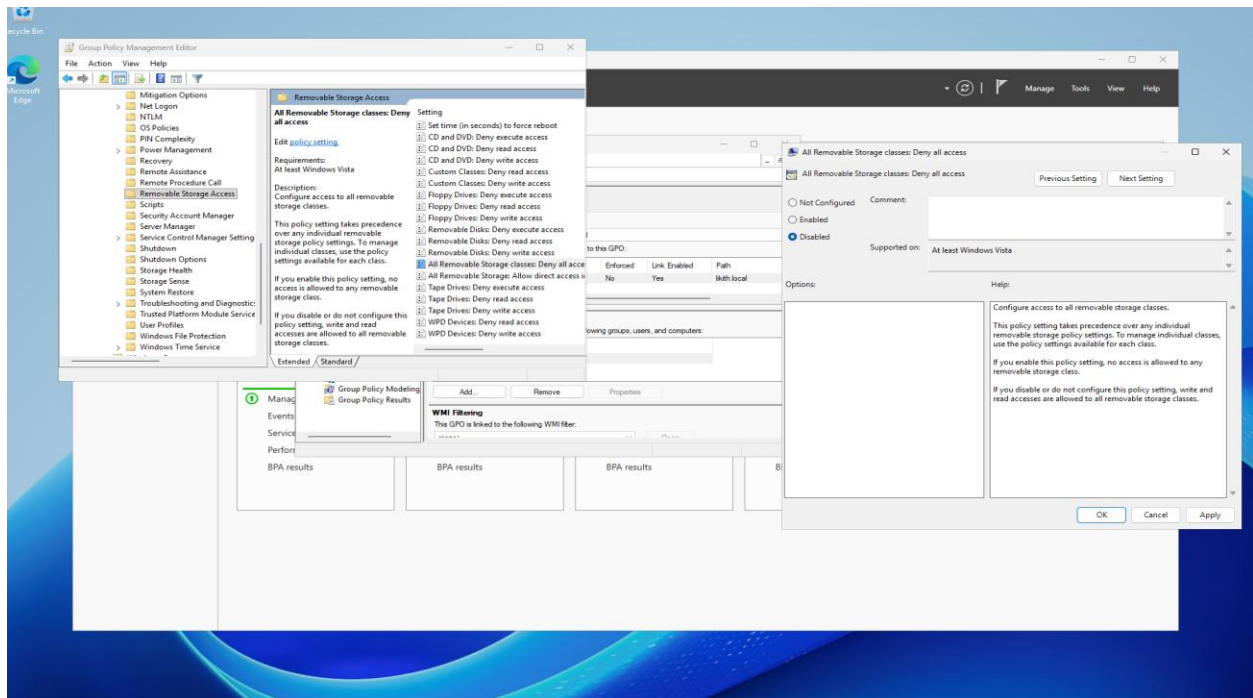
Turn on the password complexity to enable



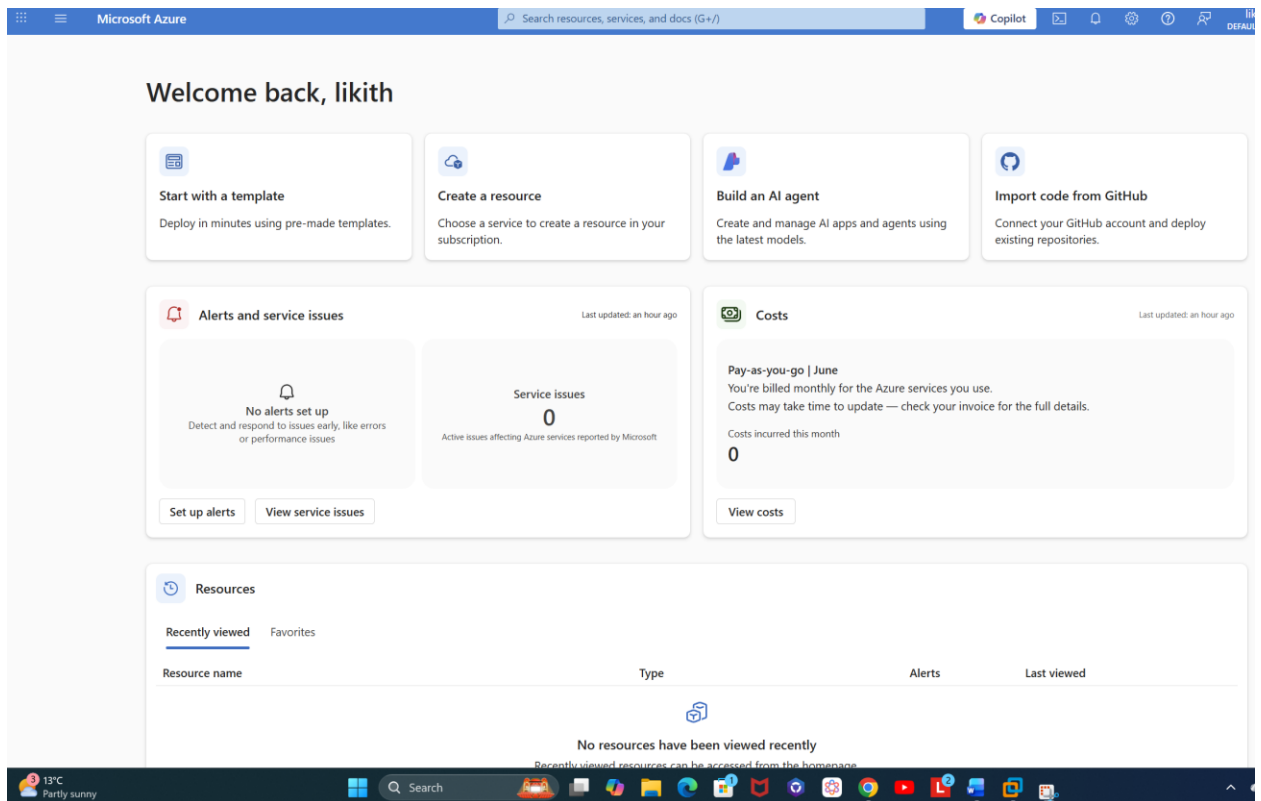
- Wallpaper Policy
Enable desktop wallpaper



- Block USB Storage
Disable USB



Step 3: Create Azure account



Verify if I global administrator role is assigned.

Download Microsoft Entra Connect from agen on the domain controller

Home

Microsoft Entra Connect | Get started ...

Microsoft Entra ID

Get started

Cloud Sync

Connect Sync

Got feedback?

Action Required: The MSOnline [deprecation](#) on April 7, 2025 will impact Entra Connect Sync service. We recommend that you upgrade your connect sync version to 2.4.18.0 or higher to avoid version. [Learn more](#)

Learn Access **Manage**

Manage your infrastructure

Determine the best solution to manage the lifecycle of your users and groups.



Manage from the cloud: Cloud Sync

Cloud Sync is the right solution for you if you are a multi-national org that wants to consolidate your identities from various disconnected Active Directories into a single Microsoft Entra tenant, or if your company is building a cloud first strategy and want to reduce the on-premises footprint. With a lightweight agent and high availability feature, Cloud Sync is the next generation synchronization tool for your needs.

Get started with Cloud Sync by downloading and installing the provisioning agent.

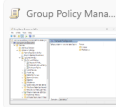
[Download Provisioning agent](#)



Manage from on-premises: Connect Sync

Microsoft Entra Connect Sync is an on-premises solution that takes care of all the operations that are related to synchronize identity data between your on-premises environment and Microsoft Entra ID. Connect Sync is used for complex topologies and by organizations that rely on their on-premises infrastructure to manage their business. Is Cloud Sync the right solution for you? [Click here to learn about Cloud Sync.](#)

[Download Connect Sync Agent](#)



Group Policy Mana...

Decline

[Accept terms & download](#)

Download Connect Sync Agent

Microsoft Entra ID

By downloading the agent, you accept our terms of service.

[Terms of Service](#)

After installing click on begin and select customize

Microsoft Entra Connect Sync

Welcome

Express Settings

Required Components

User Sign-In

Express Settings

If you have a **single** Windows Server Active Directory forest, we will do the following:

- Configure synchronization of identities in the current AD forest of **LIKITH**
- Configure password hash synchronization from on-premises AD to Microsoft Entra ID
- Start an initial synchronization
- Synchronize all attributes
- Enable Auto Upgrade

[Learn more about express settings](#)

Select Customize to choose advanced deployment options or import settings from an existing server.

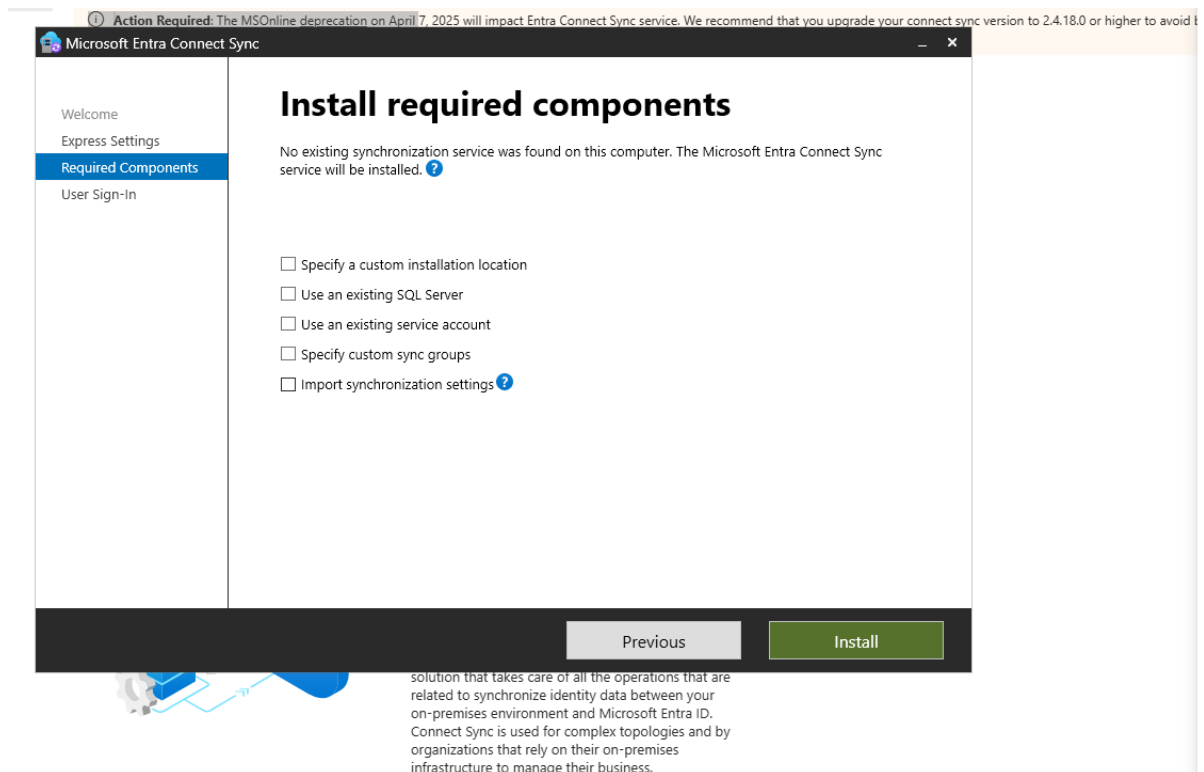
[Customize](#)

[Use express settings](#)

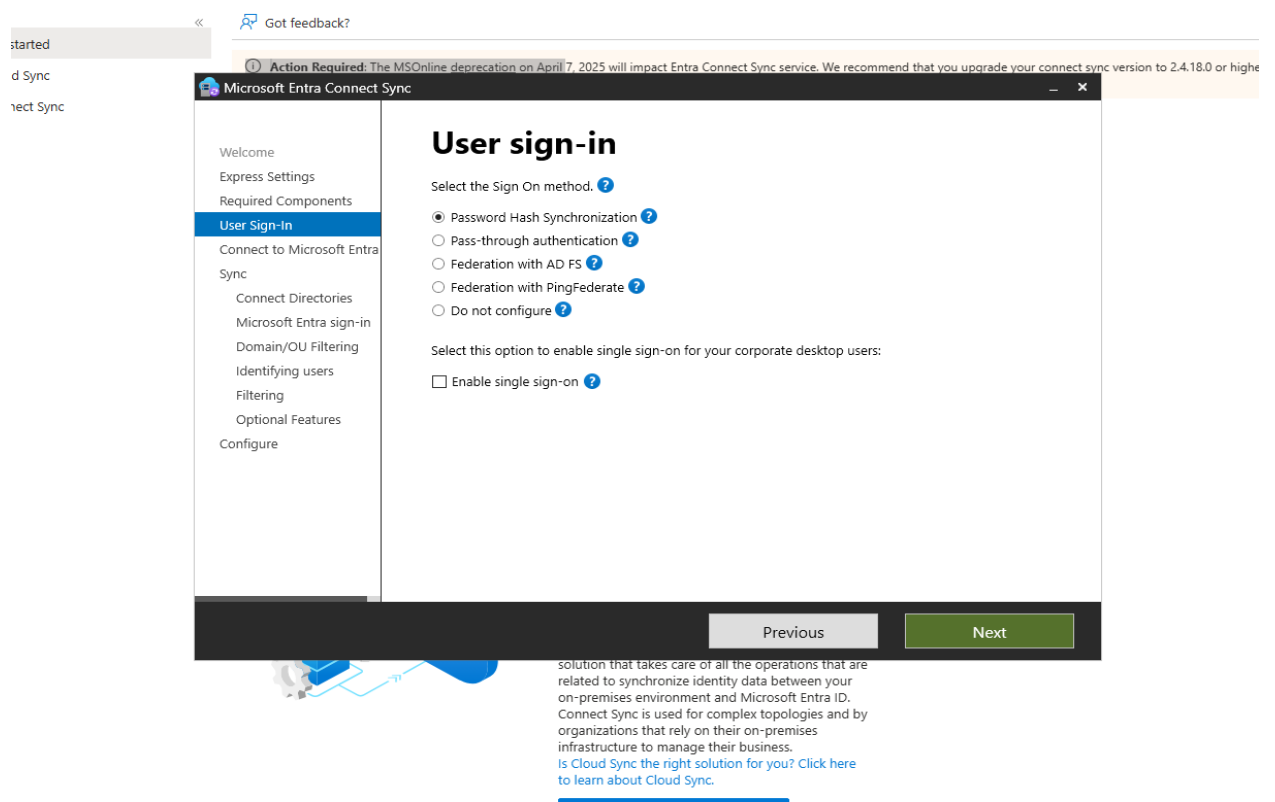
[Download Provisioning agent](#)

 **Manage from on-premises: Connect Sync**

Click install



Select password Hash synchronization



Connect to Entra ID: Enter your Azure/Entra Global Administrator account:

Enter

OverviewMonitoringProperties

Microsoft Entra Connect Sync

Express Settings

Required Components

User Sign-In

Connect to Microsoft Entra Sync

Connect Directories

Microsoft Entra sign-in

Domain/OU Filtering

Identifying users

Filtering

Optional Features

Configure

Connect your directories

Enter connection information for your on-premises directories or forests. ?

DIRECTORY TYPE

Active Directory

FOREST ?

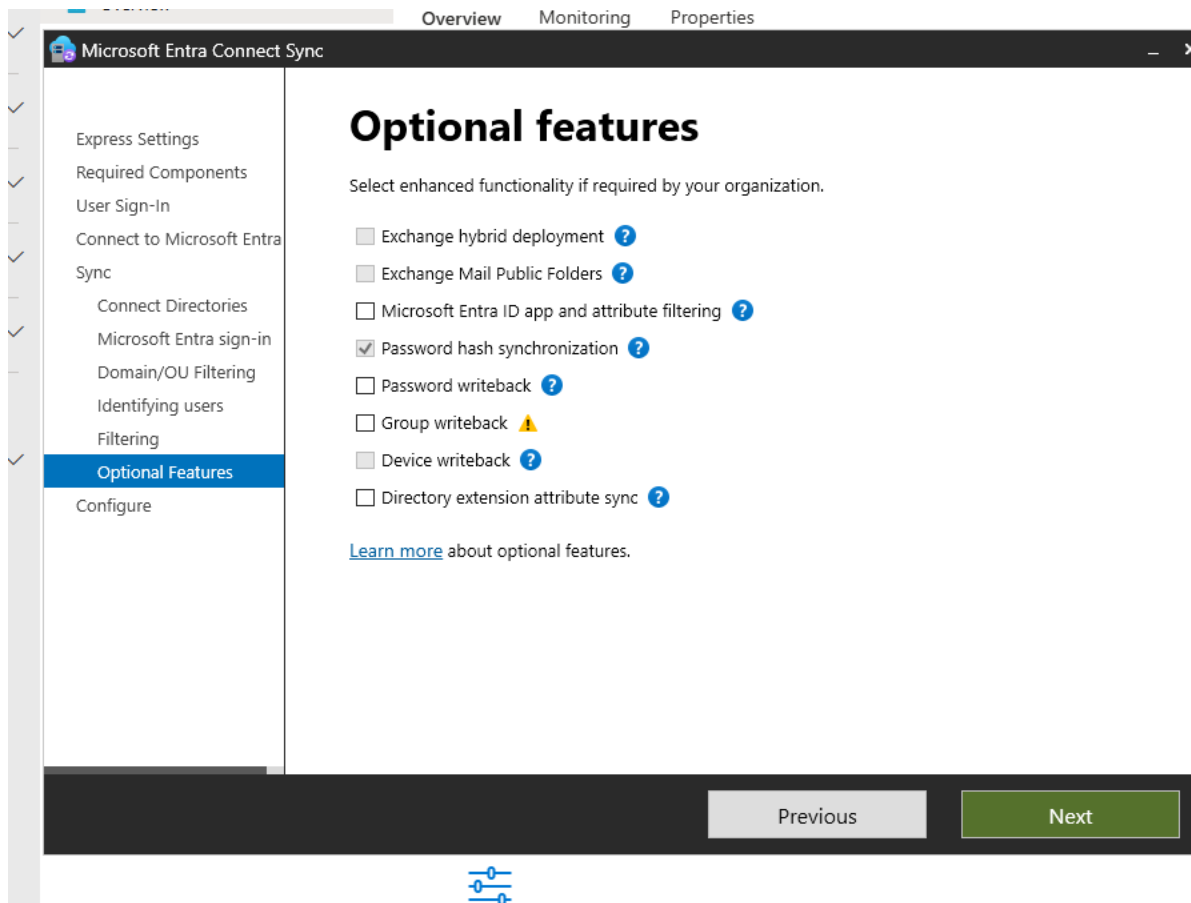
likith.local

Add Directory

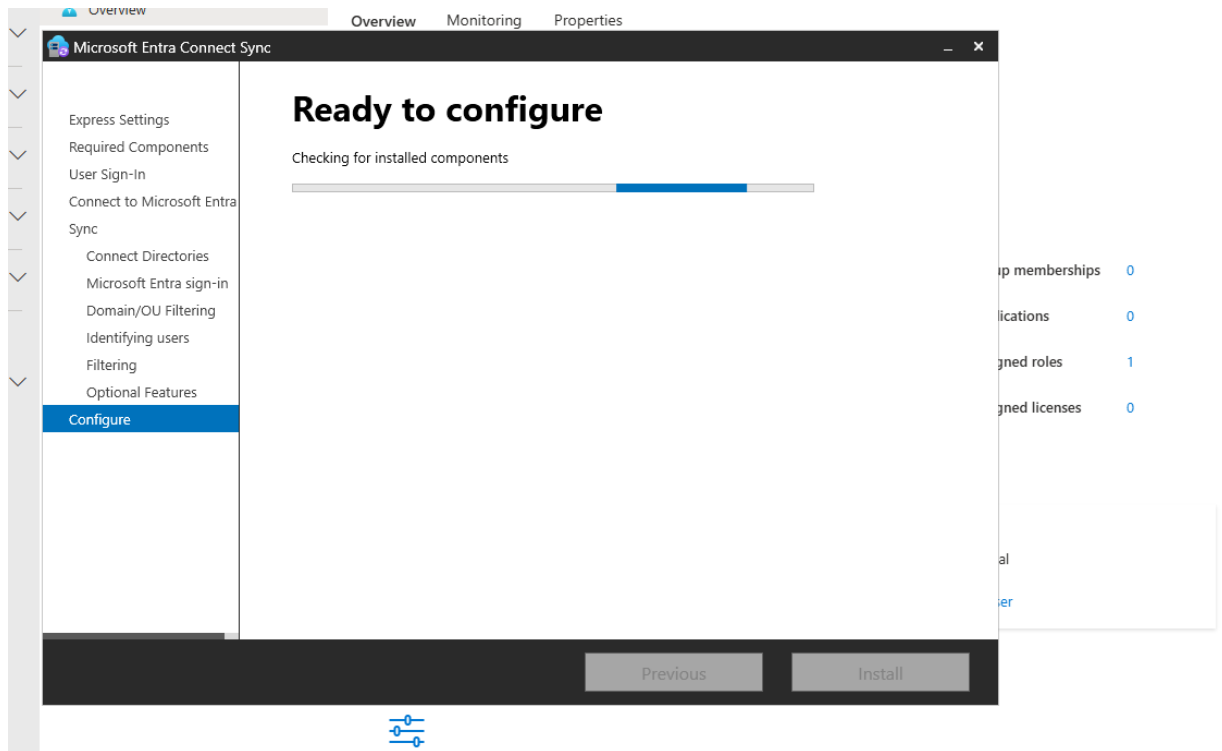
No directories are currently configured.

Previous

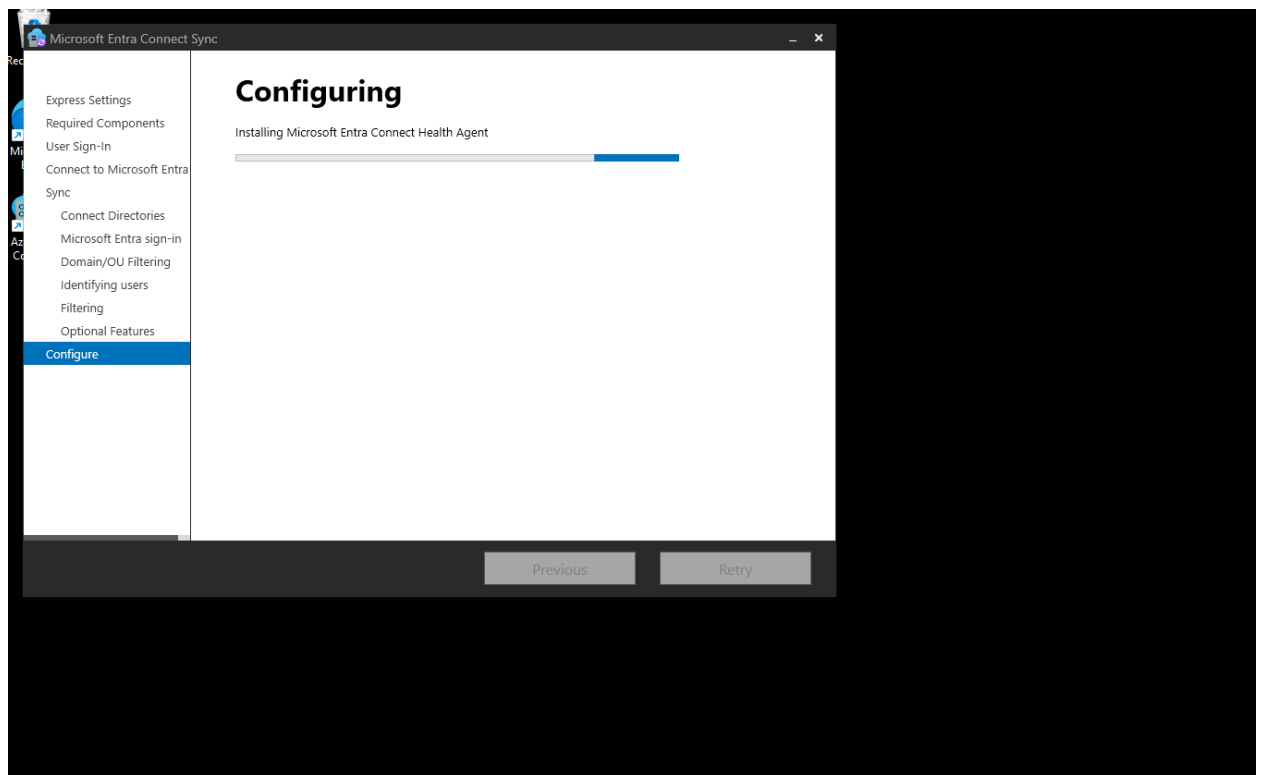
Next

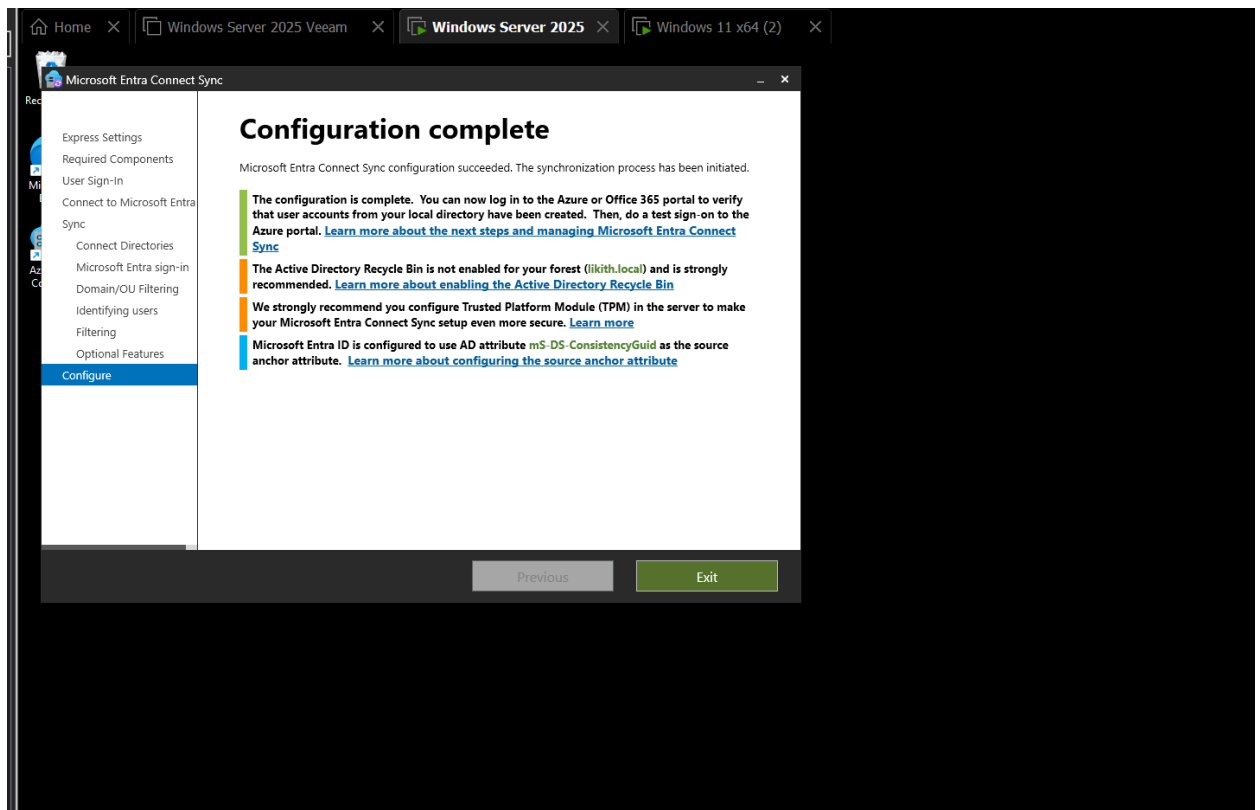


Once all the configuration is done click on Install



It will configure





Before migration:

Users Teams Products Upcoming changes (5) Learn Setup

Manage who can access apps and services included in your Microsoft 365 subscriptions. Add or remove users, manage licenses, and reset passwords.

+ Add user Reset password 1 selected Search your users list

Name ↑	Username for sign-in	Licenses
AA Adam Aiden		Unlicensed
HR Hark Rubio		Unlicensed
HJ Henry Joseph		Unlicensed
JF John Frank		Unlicensed
LU Likith U		Office 365 E5 (no Teams)
MP Miles Paul		Unlicensed
VK Vinith Kris		Office 365 E5 (no Teams)

Now we can see all the user who we on premises are migrated to the cloud

Home

Users ...

student

« + New user Edit Delete Download users (Preview) Bulk operations Refresh Manage view Per-user MFA Got feedback?

Azure Active Directory is now Microsoft Entra ID.

Search users Add filter

12 users found

☐	Display name ↑	User principal name ↑	User type	Is Agent	On-premises sy...	Identities	Company name
☐	AA Adam Aiden	[REDACTED]	Member	No	No	[REDACTED]	
☐	AJ Adam Jones	[REDACTED]	Member	No	Yes	[REDACTED]	
☐	HR Hark Rubio	[REDACTED]	Member	No	No	[REDACTED]	
☐	HJ Henry Joseph	[REDACTED]	Member	No	No	[REDACTED]	
☐	JF John Frank	[REDACTED]	Member	No	No	[REDACTED]	
☐	L likith	[REDACTED]	Member	No	Yes	[REDACTED]	
☐	LA Likith Admin	[REDACTED]	Member	No	Yes	[REDACTED]	
☐	LU Likith U	[REDACTED]	Member	No	No	[REDACTED]	
☐	ML Mike Luke	[REDACTED]	Member	No	Yes	[REDACTED]	
☐	MP Miles Paul	[REDACTED]	Member	No	No	[REDACTED]	
☐	TU Test User	[REDACTED]	Member	No	Yes	[REDACTED]	
☐	VK Vinith Kris	[REDACTED]	Member	No	No	[REDACTED]	

Final:

Test Synchronization

Create a new AD user in the Active directory:

I have created a new user as Azure user in AD.

Common Controls

- Employees
- ForeignSecurityPrincipal
- Groups
- HR

ADSyncPass... Security Group... ADSyncPasswordSet

Allowed RO... Security Group... Members in this group c...

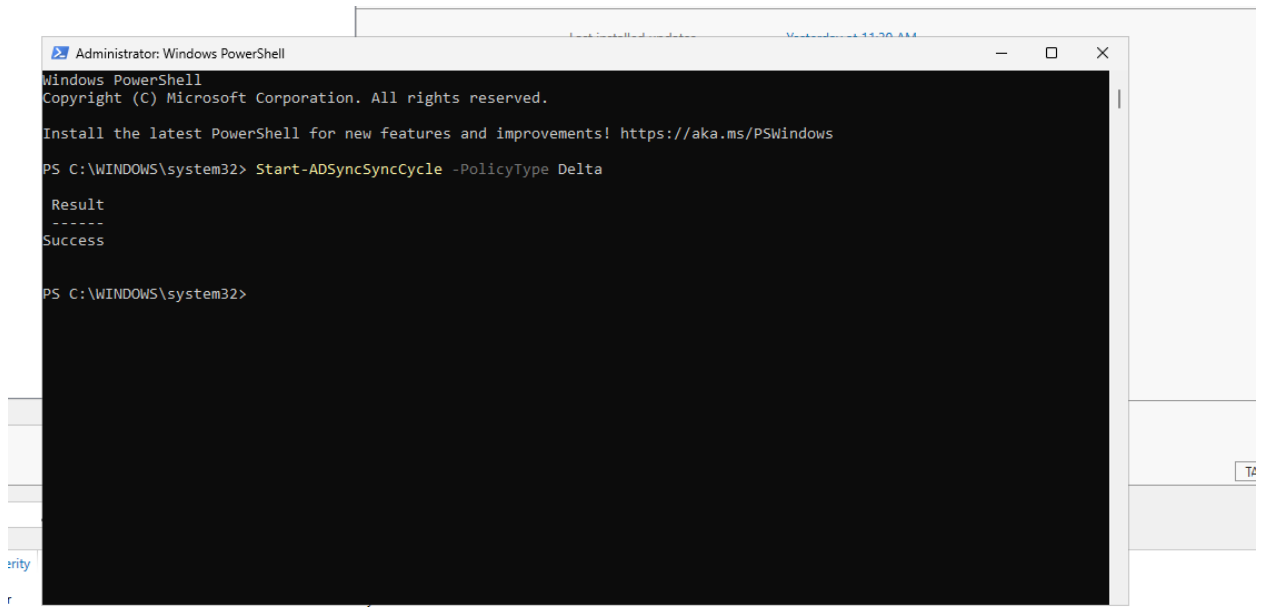
Azure User User

Cert Publish... Security Group... Members of this group ...

Then run on DC01: By default, **Microsoft Entra Connect Sync** runs every **30 minutes**. instead of waiting 30min run this command.

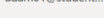
Start-ADSyncSyncCycle -PolicyType Delta

Verify the user appears in Entra ID.



r Microsoft-Windows-TPM-WMI System 6/3/2026 1:20:00 PM
ning Microsoft-Windows-Time-Service System 6/3/2026 1:15:37 PM
ning Microsoft-Windows-Windows Remote Management System 6/3/2026 1:15:08 PM
ning Microsoft-Windows-DNS Client Events System 6/3/2026 1:15:04 PM
ning Microsoft-Windows-DNS Client Events System 6/3/2026 1:14:59 PM

So we can see the user appears in Entra ID

User settings	☐	 Adam Jones			Member	No	Yes	
Bulk operation results	☐	 Azure User			Member	No	Yes	
New support request	☐	 Hark Rubio			Member	No	No	
	☐	 Henry Joseph			Member	No	No	
	☐	 John Frank			Member	No	No	
	☐	 likith			Member	No	Yes	
	☐	 likith Admin			Member	No	Yes	